

Exercise 1b - 4th of March 2024

First Clustering Task with K-Means

Deadline: 18th of March 2024 (end of day)

The aim of this exercise is to implement your first clustering task based on the MNIST dataset,¹ the same dataset you have already used for the previous classification task with KNN.

In this exercise, you will implement the **K-Means** algorithm, apply it to the training set (train.csv), and validate the quality of the clustering. The definition of the algorithms as well as the validation methods can be found in the slides.

Data and Features

The same MNIST dataset is used as in Exercise 1a. As a reminder, the csv files contain the labels (drawn number) and pixel values for the 28×28 pixels in the greyscale images, with values in the range of $[0, 255]$.

The pixel values should be used directly as the features. You don't need to compute any other features (although you certainly can if you like to).

Use the *Euclidean distance* to determine the closest centroid.

K-Means

Create your own implementation of the K-Means algorithm (see lecture notes). Determine the clusters by running it on the *training set*.

Evaluate the cluster quality with *at least two* of the following cluster validation methods:

- C-Index
- Goodman-Kruskal-Index
- Dunn-Index

¹<http://yann.lecun.com/exdb/mnist/>

- Davis-Bouldin-Index

Choose the initial centroids by randomly selecting one of the training samples. Run the clustering multiple times with different initialisations to understand its impact on the created clusters.

Expected Submission

Hand in your solutions on ILIAS `exercises/1b-KMEANS: hand-in` with:

- Source Code of your implementation.
- Short report with your results:
 - Cluster quality from *at least two* validation methods for $k = \{5, 7, 9, 10, 12, 15\}$.
 - *Optional:* Classify the test set based on the clusters and calculate the accuracy.